Jinshi's Mood Chart

Time Limit/Memory Limit: 1sec/256M

Description

Ever since Maomao accidentally made Jinshi believe that she had gone out on a date with another man, Jinshi has suffered a major emotional blow. Jinshi's elderly housekeeper then asked Maomao to keep track of his daily mood changes. She specifically warned Maomao that Jinshi's mood should not fluctuate too wildly. Ideally, the mood record should form a mountain-shaped chart. On the other hand, the chart should not stay at a low valley all day, so the total mood value should be as large as possible. The sooner Jinshi's mood can be stabilized, the sooner Maomao can stop doing this job.

Although Maomao knows the housekeeper means well, she really does not want to handle this task every day. Therefore, she asks you to write a program to solve the problem quickly.

You are given a sequence of n mood values $a_1, a_2, \ldots, a_n$. You must construct a new sequence $b_1, b_2, \ldots, b_n$ satisfying all of the following conditions:

- The new sequence must be mountain-shaped.
- The sequence must have exactly one peak; it may not contain two or more peaks.
- The peak is allowed to be at the leftmost or rightmost position.
- To avoid being discovered, you are not allowed to increase any mood value, so $b_i \le a_i$ for every i.
- Among all possible valid sequences, the sum $b_1 + b_2 + \cdots + b_n$ must be maximized.

A sequence is considered mountain-shaped if there exists an index p such that

$$b_1 \le b_2 \le \cdots \le b_p \ge b_{p+1} \ge \cdots \ge b_n,$$

and b_p is the unique maximum value in the sequence.

It is guaranteed that the optimal answer is unique. Please output the maximum possible total mood value and the resulting mood chart.

Input Format

The input consists of two lines:

- The first line contains one integer n, the number of recorded mood values.
- The second line contains n integers $a_1, a_2, \ldots, a_n$, representing the recorded mood values.

Output Format

Output two lines:

- The first line contains one integer, the maximum possible sum of the new mood values.
- The second line contains n integers, representing the new mood chart.

Adjacent integers in the second line should be separated by a single space.

Testcase Range

$1 \leq a_i \leq 10$
$1 \leq n \leq 10^3$

Sample Input/Output

Sample Input	Sample Output
5 5 3 4 1 1	13 5 3 3 1 1
6 6 5 3 9 2 7	22 3 3 3 9 2 2